
TREZ -Theory of Inventive Problem Solving

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Theory of Inventive Problem solving (TRIZ)

TRIZ ENABLES TO DO FOLLOWING

- ✓ GENERATE BETTER IDEAS FASTER
- ✓ BASIS FOR SELECTING THE BEST IDEAS
- ✓ IDEAS THAT WILL SOLVE PROBLEM & FORMS BASIS FOR FURTHER IMPROVEMENT

Examples of GOOD ideas rejected

- Antibiotic drug - Penicilline
- Fast food –McDonald
- Garbage collection & recyclable – Molok
- Flesh smelting –use of sulphur ore

The common questions

- ✓ The inventor is seldom a good salesperson
- ✓ Lack of support from management
- ✓ Poor presentation of idea
- ✓ Prejudice (Paradigm Paralysis)

Good opportunities & great losses for business

- ✓ Good ideas are rejected and resources are lost to bad one (not able to know if idea was good)
- ✓ Some patterns are repeated and we can learn from the past

Common features of GOOD SOLUTION

1. Resolve contradiction

- ✓ Tread off contradiction : if something good happens something bad also happens
- ✓ Inherent contradictions : Wanting a thing that has two opposite properties (ex, hot & cold at same time)

2. Increase ideality of system

- ✓ Measure of closeness of the system to perfect system (Ideal Final Result)
- ✓ Has all benefits customer want at no additional cost and no harmful effect
- ✓ Ideality increase when system gives more what you want and less of what you don't want at lower cost and less complexity

3. Use available resources

- ✓ Energy ,material ,information

Example of resistance to new technology

Ex1. Penicilline

C: Mold was used to resolve contradiction

I: Many other diseases were cured by use of penicillin

Ex2 .Collar

C: No separate T fitting & tube used as resource

I: one joint less ,less complex ,material & labor

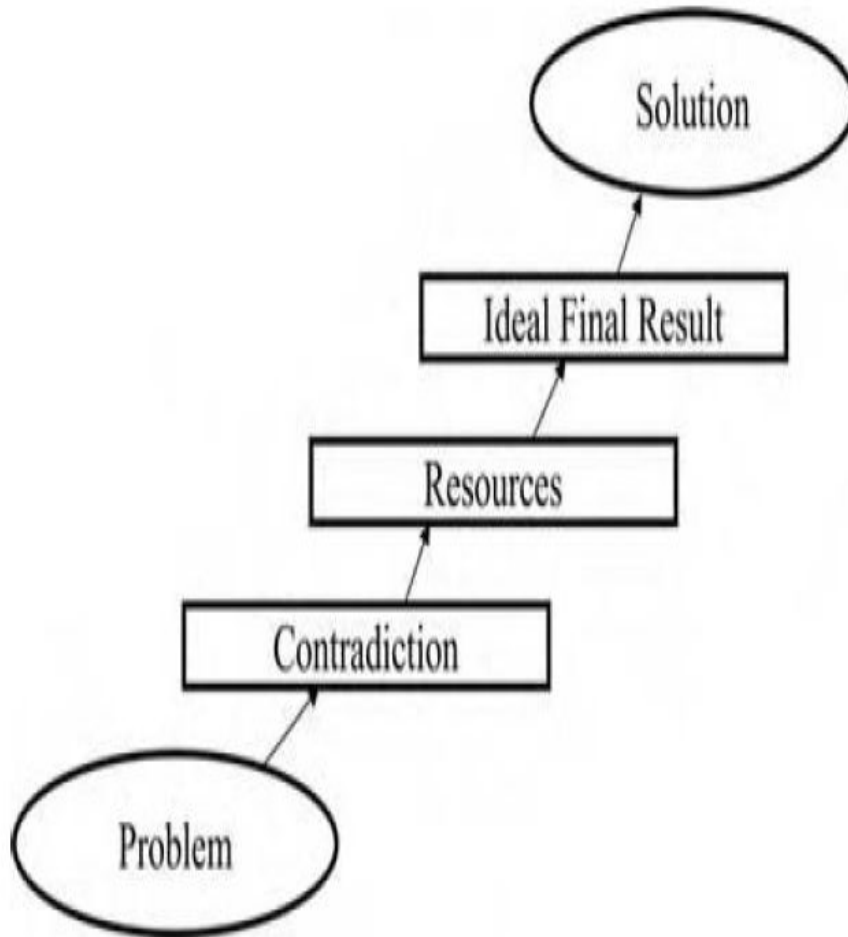
Ex3: Fast Food

C: No waiters at same time customer as their own waiters

R: Customers

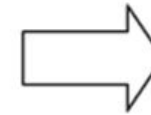
I: better & quick service

Features of Good solution



Old way of working

Trial and error ✓
Contradictions are hidden ✓
Adding resources ✓
Compromise ✓
Evolution is considered accidental ✓



New way of working

Systematic work ✓
Contradictions are clarified ✓
Using available resources ✓
Ideal Final Result
Knowledge of the patterns of evolution ✓

Construction of TRIZ model with 5 steps

1. Describe concept of contradiction
2. Mapping resource addition to model
3. Concept of ideal final result is formulated
4. Pattern of evolution
5. Innovative principles

1. Contradiction (C)

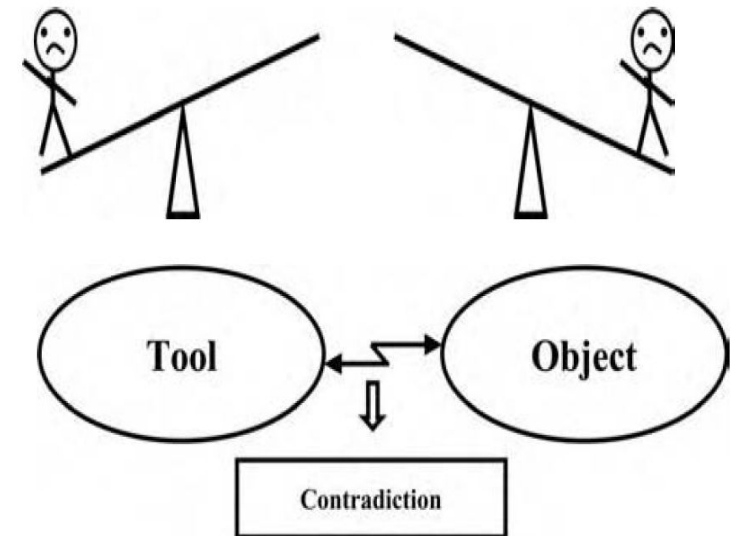
system consists of two components: tool (T) and object (O).

Example :

1. Ax blade (T) Wooden block (O)
2. Training (T) Professional quality of employee (O)

C : the better the service, the more training time is needed

3. Sea saw system



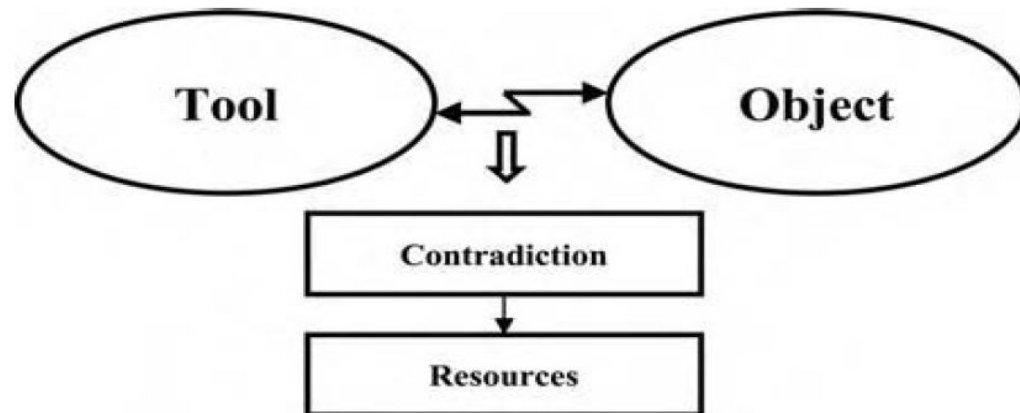
2. Resources

Things, information, energy, or properties of the materials that are already in or near the environment of the problem

They can be added directly or modified to solve the problem (eliminate contradiction)

Examples

1. Garbage bin
2. Ax problem (resources are the blade, its edge, form, material, and other properties)
3. Training example



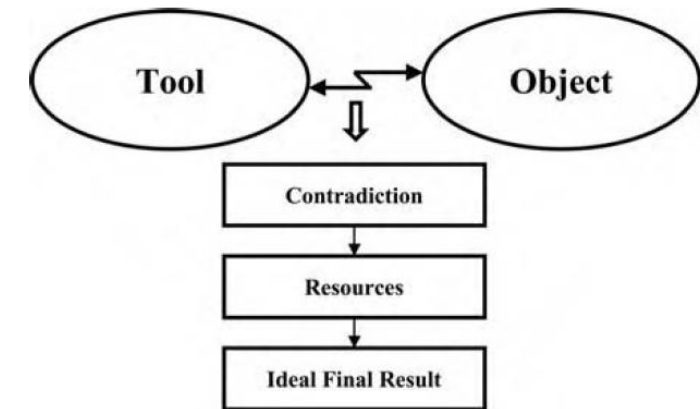
3. Ideal Final Result

Example Ax problem

blow struck by the ax should be made stronger, but, at the same time, the tool should remain easy to use.

To split the chunk of wood, the ax should be heavy, but, at the same time, for ease of handling, the ax should be light.

Problem solved by using hollow handle



The ideal final result is the solution that resolves the contradiction without compromise.

Resources are used to go from the contradiction to as perfect a solution as possible.

4. Pattern of Evolution

Methods to follow to resolve contradictions, to use resources, and to make the system more ideal

- ❖ Features : any properties of systems: size, weight, speed, flexibility, color ...
- ❖ Patterns of evolution are important regularities in the development

□ Transition from the macro- to the microlevel or the division of the system to smaller parts.

✓ Electronics

✓ Microwaves

✓ Beds (water ,air neds) & mattress

Different parts of a system may change at different rates and may follow different patterns

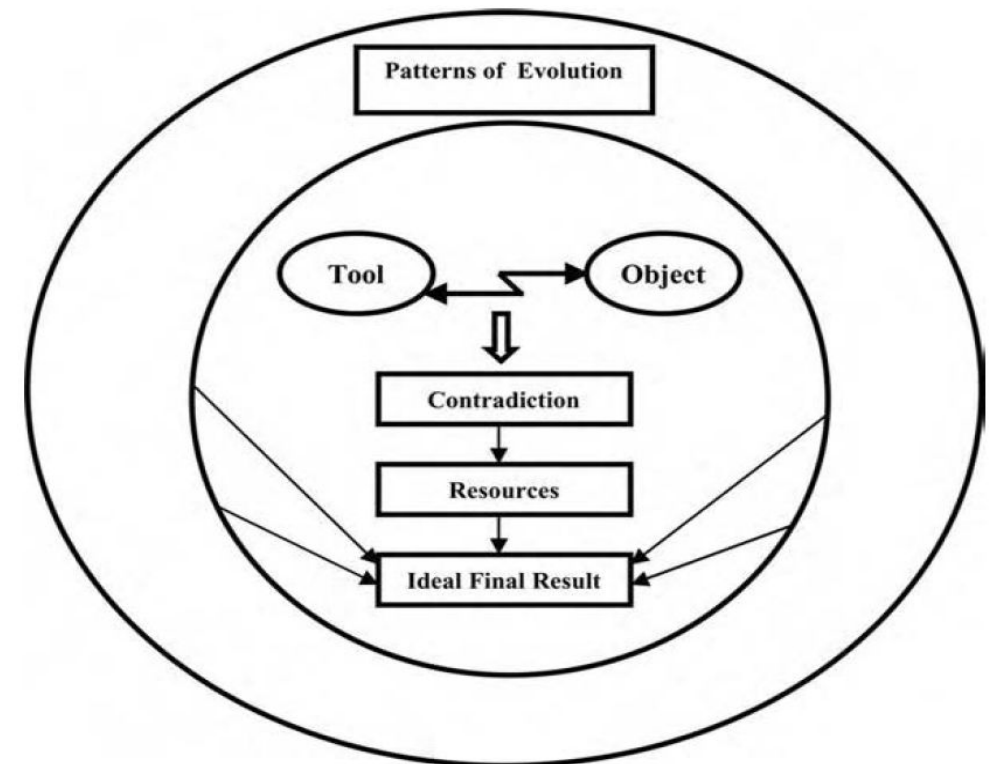


□ Transition to Macrolevel

The parts of the system become more interactive with each other. The system is expanded and convoluted.

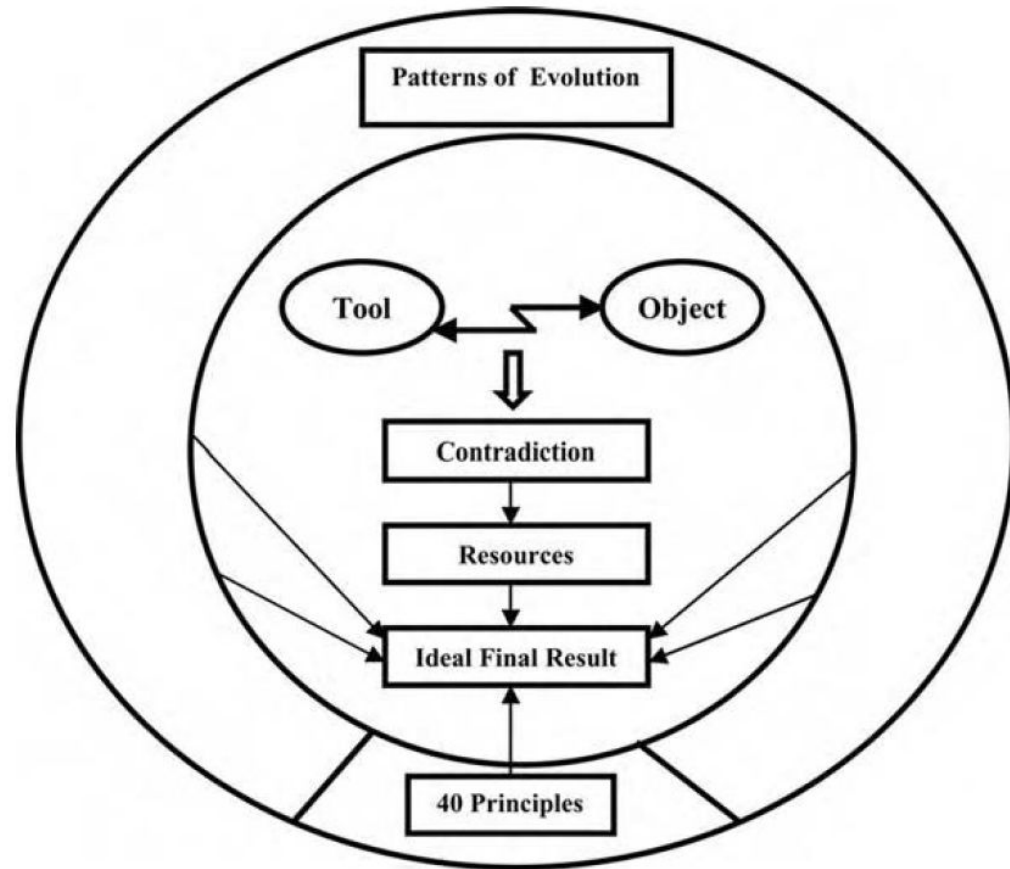
Addition of new features

combining all the features into a new, simpler system that has all the benefits without all the complexity



5. Innovative principles

Are tools that tell what the patterns mean and help us interpret the patterns for any particular problem.



Trade off

something good happens, something bad happens.
Alternatively, when

something good gets better, something undesirable gets worse

Examples

The product gets stronger (good) but the weight increases (bad).

Software is made easier to use (good) but versatility decreases (bad).

The hot coffee is enjoyable to drink (good) but can burn the customer (bad).

Training gets more thorough (good) but requires more time (bad).

The faster the automobile airbag deploys, the better it protects the occupant (good), but the more likely it is to injure or kill small people or out-of-position people (bad).

Inherent Contradictions

There are always inherent contradictions behind tradeoffs—sometimes they are obvious and sometimes they are hidden.

Examples

The product should be thick (to get needed strength) yet should be thin (to be light).

Software should have very few options for ease of use and should have numerous options to be effective.

Coffee should be hot for enjoyable drinking and should be cold to prevent burning the customer.

Training should be lengthy to ensure good learning and should be very short to minimize demands on time.

The automobile airbag should deploy quickly to save the driver or passenger, yet should deploy slowly to minimize harm to small drivers or passengers.

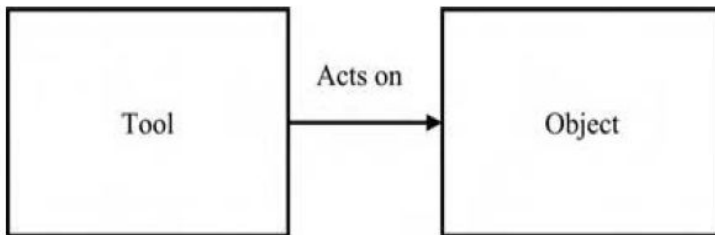
Defining Trade off

The tradeoff arises in a system consisting of a tool and an object

Tools

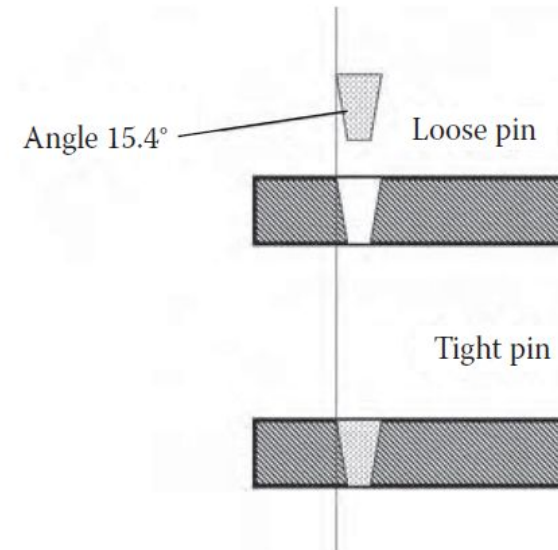
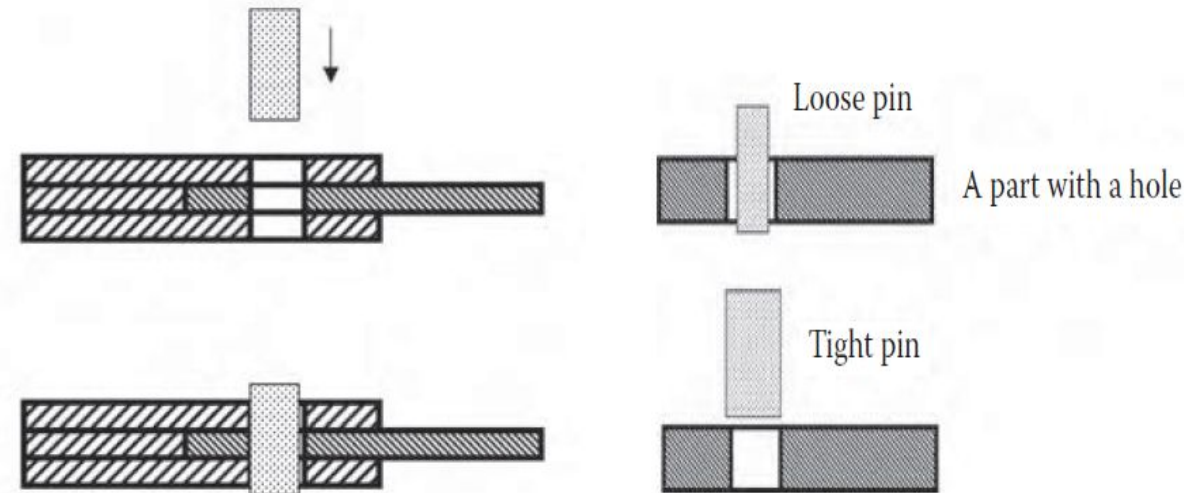
- ✓ Scissors ,transisters
- ✓ Knowledge, models, an d information are also tools

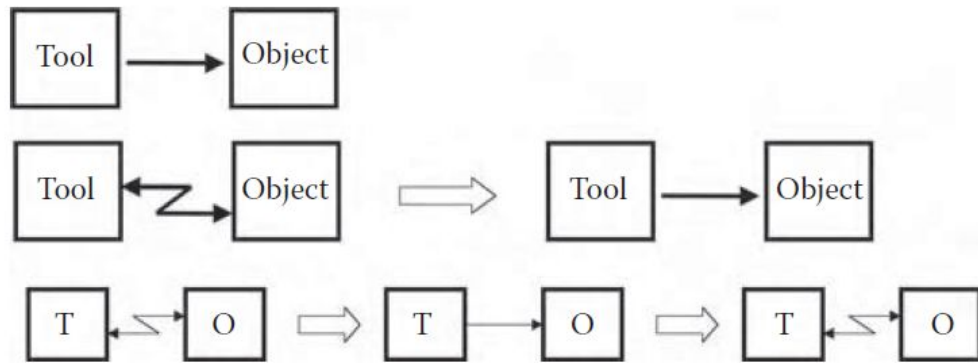
The tool is the component that is easiest to change when the problem is resolved.



Objects

The object, on the other hand, gives constraints for the change of the tool





tradeoffs in the system even when there are no visible harmful effects.

Ex

1. The electric car is silent and does not produce exhaust gases, but the power system occupies a great deal of the space in the car and reduces the load it can carry

tradeoff may appear clearly as a harmful action that accompanies a useful action

Ex

1. The car transports people and cargo, but the noise of the car disturbs people and exhaust gases pollute the atmosphere.

2. The transistor produces harmful heat while it does beneficial switching.

Different Models of features

Features and actions can describe tradeoffs.

the development of a product means the improvement of the features

Ex : features of

1. Muffler in the conventional lawnmower

- ✓ Dimensions
- ✓ Weight
- ✓ Noise absorption capacity
- ✓ Ease of manufacture
- ✓ Form, outer appearance

2. pin in latch

Dimensions

Form

Surface quality, manufacturing precision

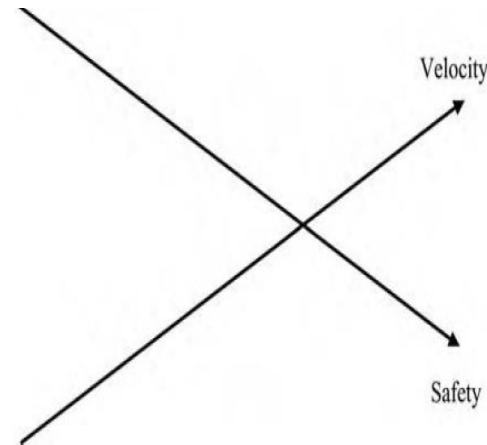
Ease of manufacture

Ease of locking and opening

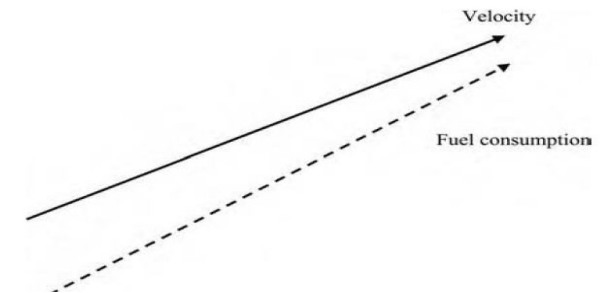
Reliability

The tradeoff between tool and object shows where the problem lies.

Its important to express the tradeoff as a conflict between two features

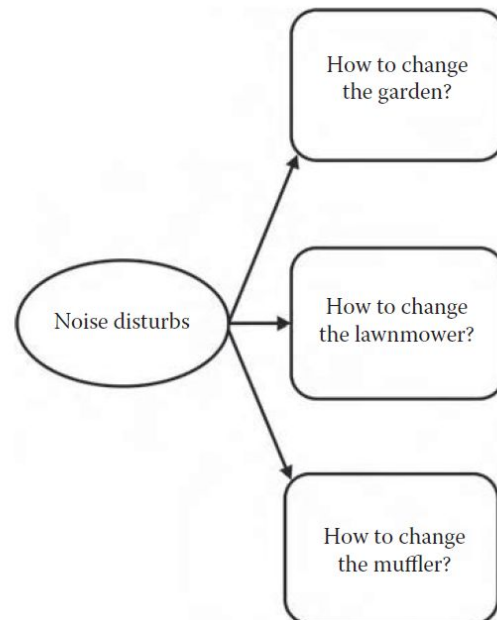
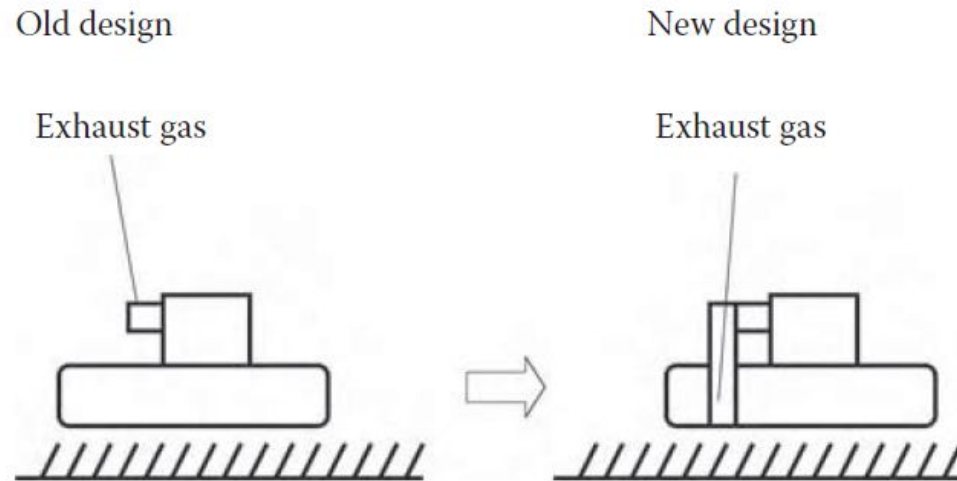


A conflict between two useful features.



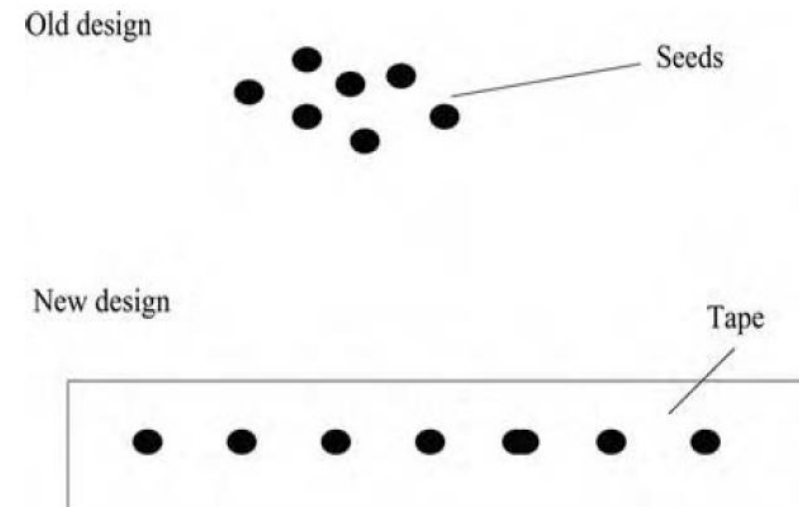
Dotted line indicate harmful features

Where does trade off appear ?



When does trade off appear ?

- Problem of Carrot cultivation
 - ✓ Seeding – if done properly thinning not needed .It can be mechanized but it involves machine ,cost ,maintenance
 - ✓ Thinning - mechanized system will involve cost ,maintenance
 - In this case we can improve seeding process (use of biodegradable tape) and eliminate thinning



Nine screen

location of the problem in time and on the system level can be illustrated by a simple table containing nine screens

	Past	Present	Future
Macro-level		Lawn-mower	
System		Muffler	
Micro-level		Porous substance	

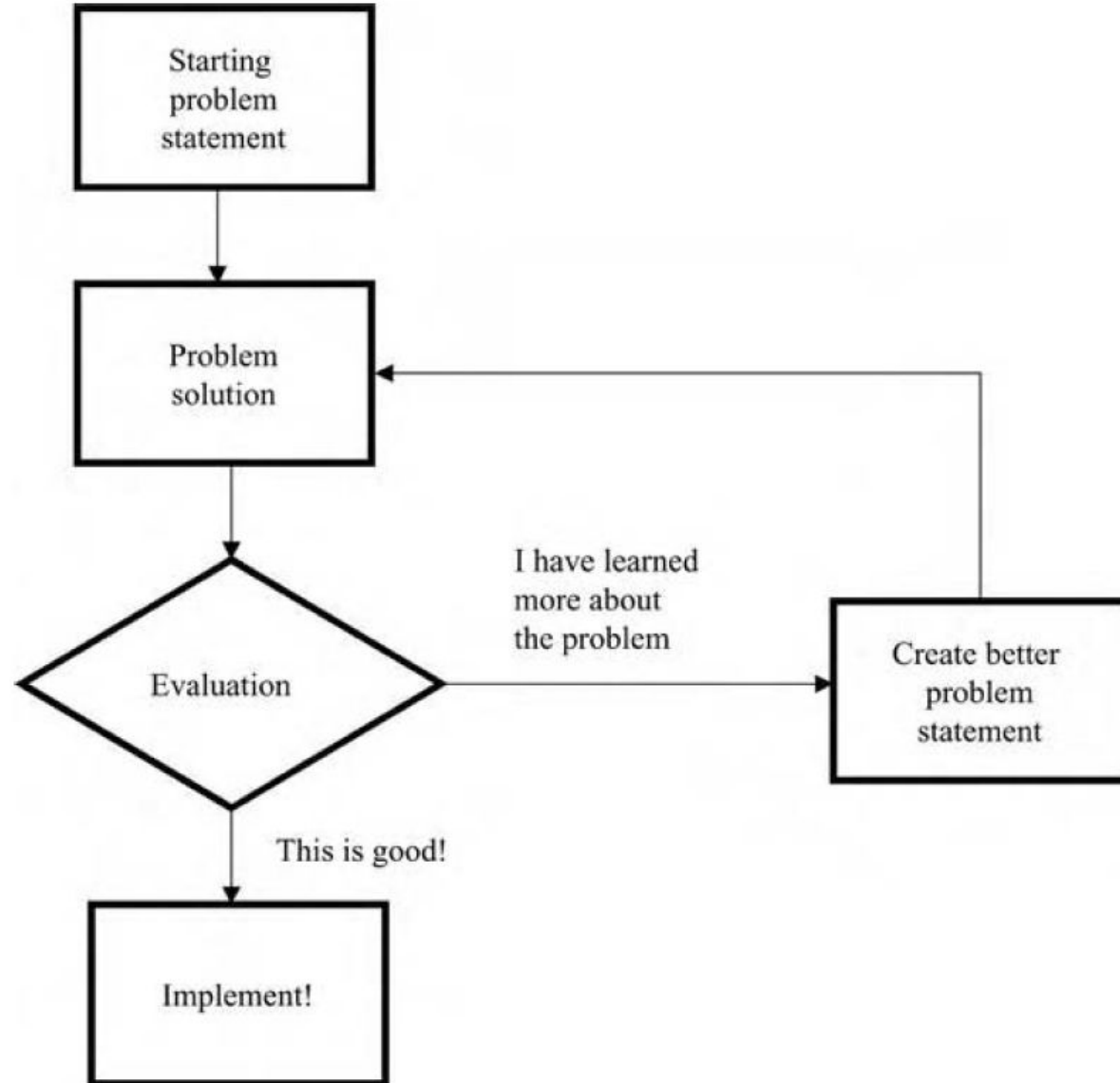
	Past	Present	Future
Macro-level	Use of tape to carry out seeding and can be improved to use the fertilizers too		
System	Carrot: seeding	Carrot: thinning	Carrot: consumption
Micro-level			

How to decide to develop or remove it ?

What are the good and bad features of this component or operation? What gets better and what gets worse, if we leave this out?"

Examples

1. **Spare Tire** : Improves reliability but also occupies space. But now have tubeless tires
2. **T- fitting joint** in pipes replaced by collared pipe
3. **In harvesting** : cutting and reaping can be done same time
4. Digital printing removes typesetting
5. Storage & warehousing elimination of electronic equipment's by manufacturing on demand and delivering



Construction of a model of trade-off : Lawnmower example

1.	Describe pairs of tools and objects.
a.	Muffler and noise (or air that vibrates)
b.	Noise and a person (disturbed by noise)
c.	Engine and exhaust gas (engine moves gas)
2.	Select one pair. Explain why just this tool and object are selected.
	We select the pair "muffler and noise" because we want to select a limited problem. We want remove the harmful feature with minimal changes in the system.
3.	Describe features of the selected system of a tool and an object. Describe conflicts in this system.
	Features: dimensions, weight, absorption capacity, ease of manufacture, form, etc.
	Conflicts between features:
a.	When noise absorption improves (+), the dimensions of a muffler increase (–)
b.	When noise is suppressed (+), the number of parts increases (muffler needed) (–)

4.	Select one pair of conflicting features. Explain why just this tradeoff is selected. We select the formulation: "When noise is suppressed (+), the number of parts increases (muffler needed) (–)."
	The reason: A limited problem, very big changes not needed, at the same time the goal is more than simple optimization of the size.
5.	Describe the tradeoff graphically and in words.
	Graphically: see Figure 3.17.
	In words: When noise absorption capacity improves, dimensions (of the muffler) and number of parts (the muffler is needed) are increased, too. If the system is simplified (or muffler made smaller or removed totally), the noise absorption capacity is lessened or lost.

Inherent contradiction

- A contradiction may not just exist between tool & object but may be with in objects of the system

To act on the object, the tool should have certain weight, dimensions, energy consumption, and other features

At the same time, we want to minimize weight, size, and energy losses

Examples of conflicts :

1. Big–small: To be comfortable, the house should be big and, to decrease the consumption of material and energy, the same house should be small.

2. Heavy–light: Strength, reliability, and safety require that a vehicle should be heavy, but energy economy requires that the same vehicle should be light

3. High energy consumption–low energy consumption: Machinery and tools need to use energy to work on objects, and they should use as little energy as possible